All for one and none forall

Compiling polymorphic relations without monomorphization

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miniKanren vs. semiringKanren

miniKanren

- evaluation via stream search (top-down)
- dyamically typed (s-expressions)
- relations are unweighted (usually)

semiringKanren

- evaluation via multidimensional array operations (bottom-up)
- statically typed (finite algebraic data types)
- relations are weighted

semiringKanren represents relations as finite multidimensional arrays.

Polymorphic relations need to represent arbitrarily many specifically-typed relations.

How to do both?

All for one and none forall

Represent all concrete instances of a polymorphic relation

using only a single concrete instance

to compile away type variables

Representing relations


```
(defrel (graph
  (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

from / to

	0	1	2	3
0	0	1	0	0
1	1	0	0	0
2	0	1	0	1
3	0	0	0	0

Represent graphs as relations or as adjacency matrices.
semiringKanren computes matrices for arbitrary relations!

Primitive relations

(== x y)

x \ y	0	1	2	3
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1

(=/= x y)

x \ y	0	1	2	3
0	0	1	1	1
1	1	0	1	1
2	1	1	0	1
3	1	1	1	0

`disj` is addition

```
(defrel (r0 (x : (Num 4)) (y : (Num 4)))
  (disj
    (== x 0)
    (== y 0)))
```

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

conj is multiplication

```
(defrel (r1 (x : (Num 4)) (y : (Num 4)))
  (conj
    (== x 0)
    (== y 0)))
```

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

`fresh` is summation

Vertices with incoming edges.


```
(defrel (incoming-edge-vertices (y : (Num 4)))
  (fresh (x : (Num 4))
    (graph x y)))
```

$$\sum_{x} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix}$$

Types

semiringKanren values inhabit finite algebraic data types.

			left		right		
`(left 0)`	:	`(Sum (Num 2) (Num 3))`	1	0	0	0	0
`(left 1)`	:	`(Sum (Num 2) (Num 3))`	0	1	0	0	0
`(right 0)`	:	`(Sum (Num 2) (Num 3))`	0	0	1	0	0
`(right 1)`	:	`(Sum (Num 2) (Num 3))`	0	0	0	1	0
`(right 2)`	:	`(Sum (Num 2) (Num 3))`	0	0	0	0	1

(semiringKanren also supports unit and product types)

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} * \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Computing relations

```
(defrel (graph (from : (Num 4)) (to : (Num 4)))
  (disj
    (conj (== from 0) (== to 1))
    (conj (== from 1) (== to 0))
    (conj (== from 2) (== to 1))
    (conj (== from 2) (== to 3))))
```

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$x \mapsto$ (left 0)

$y \mapsto$ (right 0)

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 & 0 & 0 \end{bmatrix}$$

$x \mapsto$ (right 2)

$y \mapsto$ (left 2)

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Sum swap

```
(defrel (sum-swap-3-3
  (x : (Sum (Num 3) (Num 3)))
  (y : (Sum (Num 3) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 3))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Larger sum swap

```
(defrel (sum-swap-3-4
  (x : (Sum (Num 3) (Num 4)))
  (y : (Sum (Num 4) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 4))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Larger sum swap

```
(defrel (sum-swap-3-4
  (x : (Sum (Num 3) (Num 4)))
  (y : (Sum (Num 4) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 4))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$x \mapsto$ `(left 0)`

$y \mapsto$ `(right 0)`

Larger sum swap

```
(defrel (sum-swap-3-4
  (x : (Sum (Num 3) (Num 4)))
  (y : (Sum (Num 4) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 4))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & 0 & 0 & 0 \end{bmatrix}$$

$x \mapsto$ (right 3)

$y \mapsto$ (left 3)

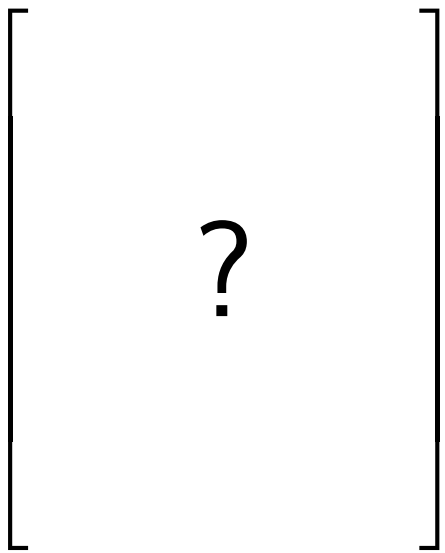
Larger sum swap

```
(defrel (sum-swap-3-4
  (x : (Sum (Num 3) (Num 4)))
  (y : (Sum (Num 4) (Num 3))))
  (disj
    (fresh (a : (Num 3))
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : (Num 4))
      (conj
        (== x (right b))
        (== y (left b))))))
```

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Polymorphic sum swap

```
(defrel (sum-swap
  (x : (Sum α β))
  (y : (Sum β α)))
  (disj
    (fresh (a : α)
      (conj
        (== x (left a))
        (== y (right a))))
    (fresh (b : β)
      (conj
        (== x (right b))
        (== y (left b))))))
```



All for one and none forall

Represent all concrete instances of a polymorphic relation

using only a single concrete instance

to compile away type variables

Expansion

semiringKanren computes a small array, and *expands* it.

Each expanded array entry adopts weight from a smaller array entry with the same *equality pattern*.

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Fails for:

$$x \mapsto (\texttt{left}\ \__\alpha)$$
$$y \mapsto (\texttt{left}\ \__\beta)$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Fails for:

$$x \mapsto (\texttt{right}\ \__\beta)$$
$$y \mapsto (\texttt{right}\ \__\alpha)$$

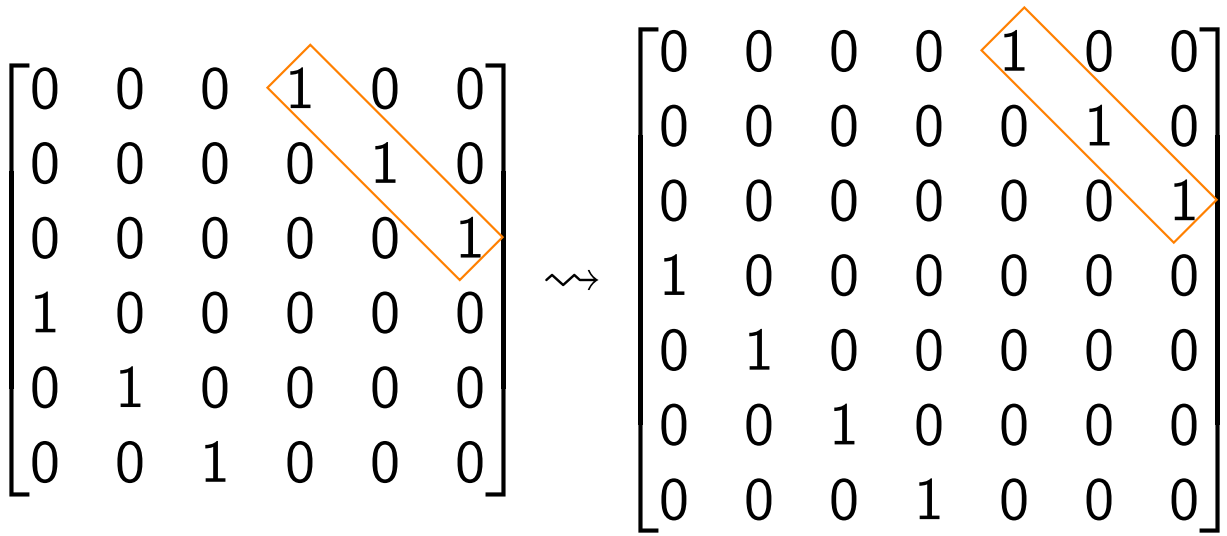
Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & \boxed{1} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Succeeds for:

$$\left.\begin{array}{r} x \mapsto (\texttt{left}\ \underline{0}) \\ y \mapsto (\texttt{right}\ \underline{0}) \end{array}\right\} \alpha\text{-typed parts equal}$$

Expanding sum swap



Succeeds for:

$$\left.\begin{array}{r} x \mapsto (\texttt{left}\ \__\alpha) \\ y \mapsto (\texttt{right}\ \__\alpha) \end{array}\right\} \alpha\text{-typed parts equal}$$

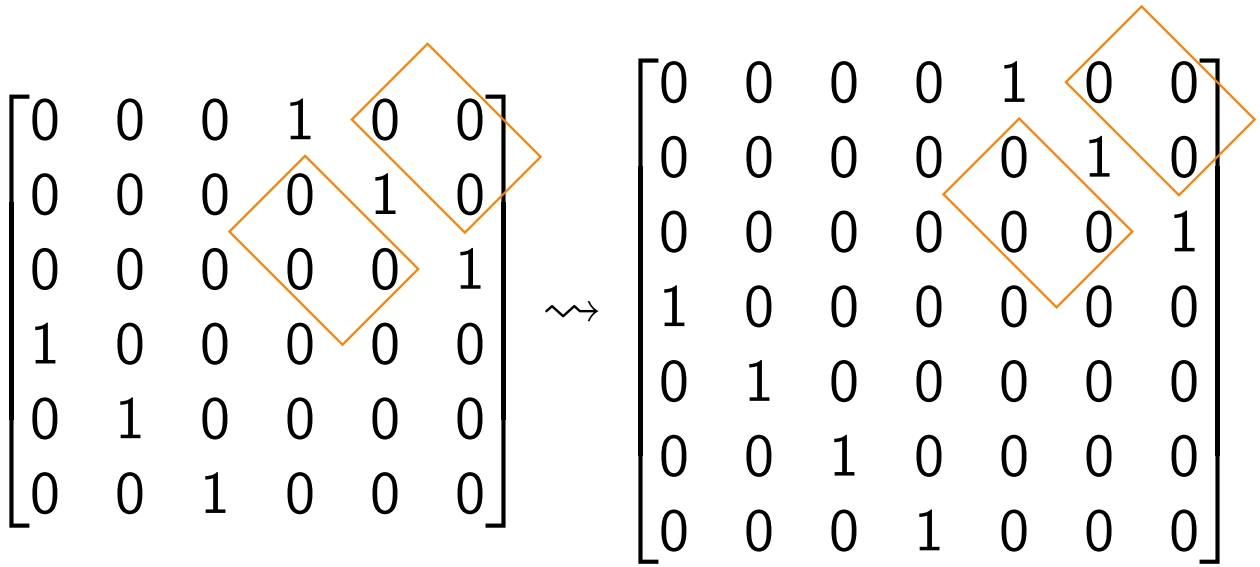
Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & \boxed{0} \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & \boxed{0} \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Fails for:

$$\left.\begin{aligned} x &\mapsto (\texttt{left}\ \underline{0}) \\ y &\mapsto (\texttt{right}\ \underline{2}) \end{aligned}\right\} \alpha\text{-typed parts disequal}$$

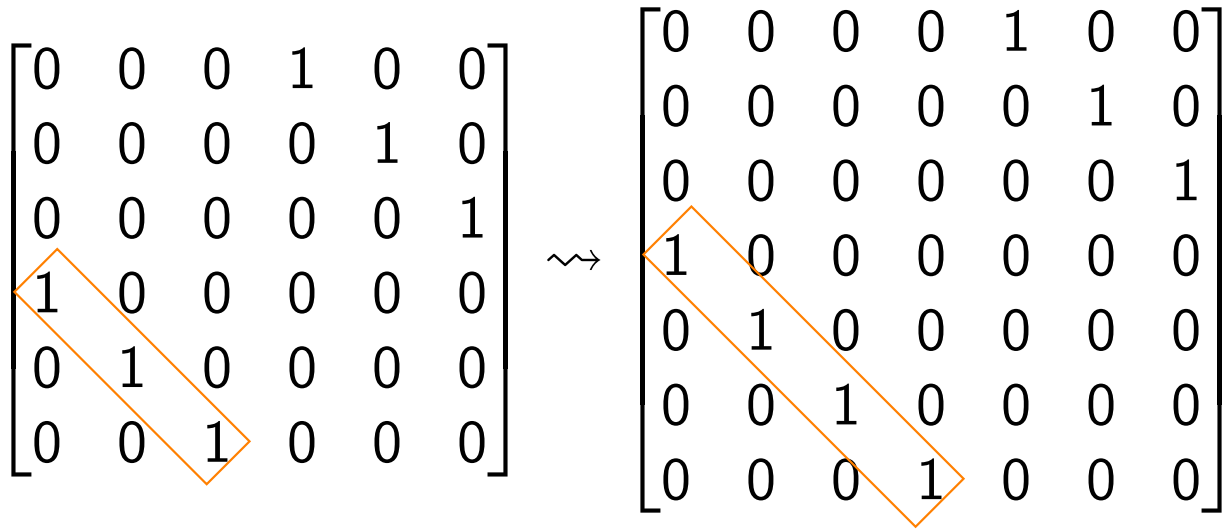
Expanding sum swap



Fails for:

$$\left.\begin{array}{r} x \mapsto (\texttt{left}\ \ \__\alpha) \\ y \mapsto (\texttt{right}\ \ \__\alpha) \end{array}\right\} \alpha\text{-typed parts disequal}$$

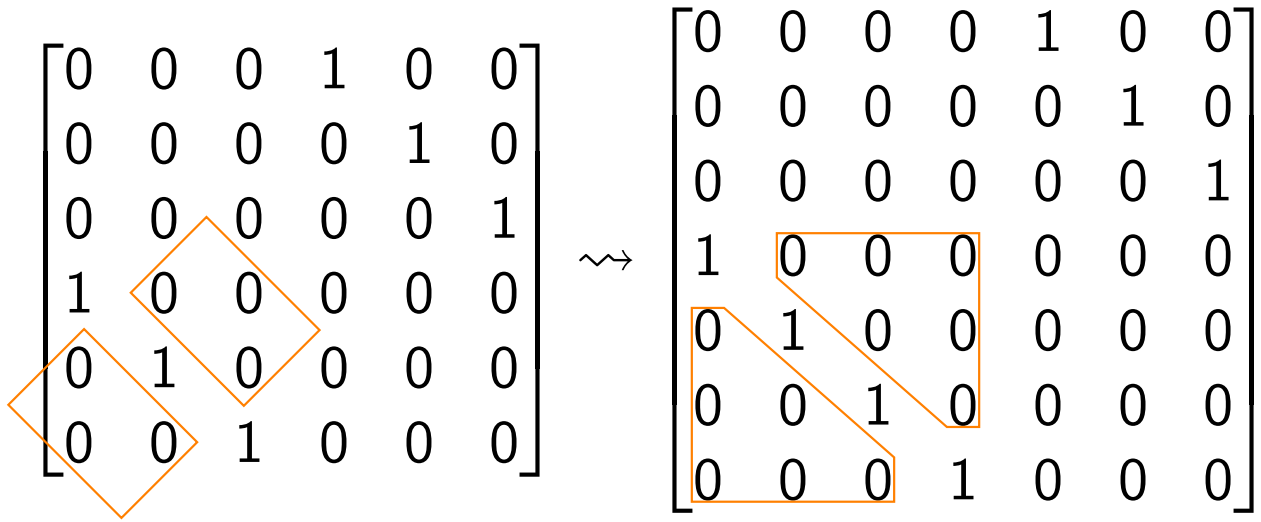
Expanding sum swap



Succeeds for:

$$\left.\begin{array}{r} x \mapsto (\texttt{right}\ \__\beta) \\ y \mapsto (\texttt{left}\ \__\beta) \end{array}\right\} \beta\text{-typed parts equal}$$

Expanding sum swap



Fails for:

$$\left.\begin{aligned} x &\mapsto (\texttt{right}\ \__\beta) \\ y &\mapsto (\texttt{left}\ \__\beta) \end{aligned}\right\} \beta\text{-typed parts disequal}$$

Expanding sum swap

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

Same success/failure results when:

- Non-variable typed parts are the same
- Variable-typed parts have the same (dis)equalities

Equality patterns

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \boxed{1} & 0 & 0 & 0 \end{bmatrix}$$

$x \mapsto (\texttt{right } \underline{2}_\beta)$

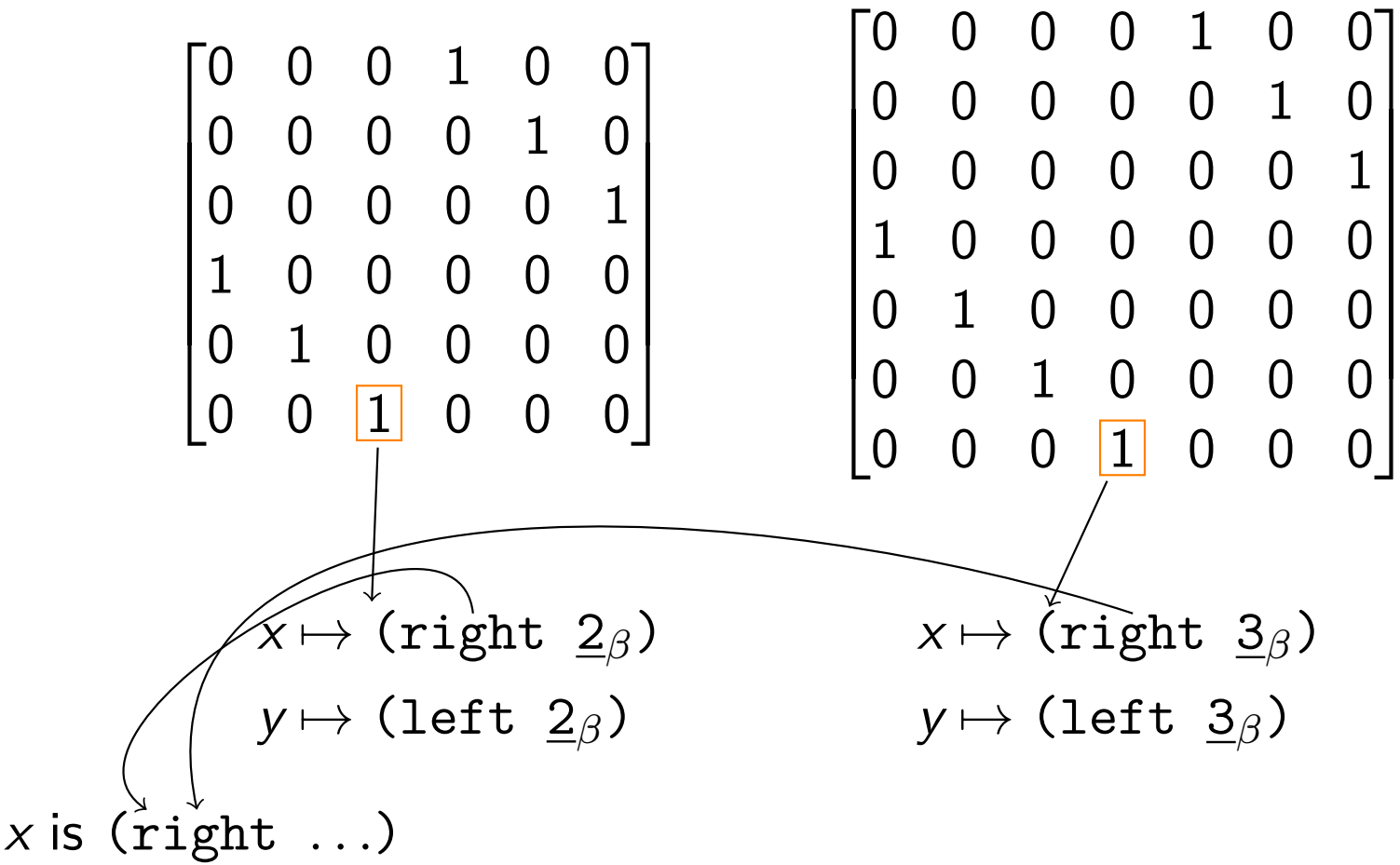
$y \mapsto (\texttt{left } \underline{2}_\beta)$

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & 0 & 0 & 0 \end{bmatrix}$$

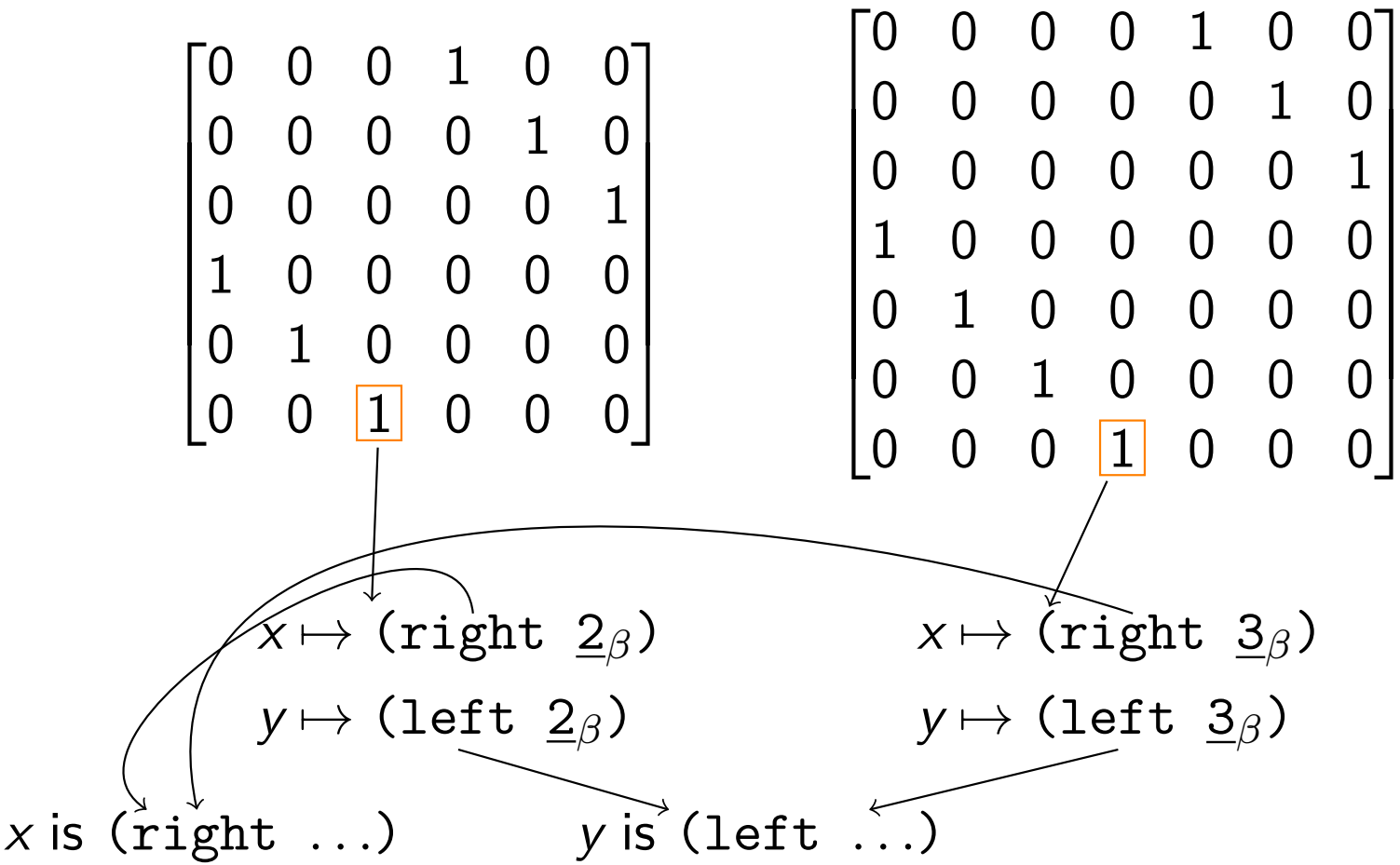
$x \mapsto (\texttt{right } \underline{3}_\beta)$

$y \mapsto (\texttt{left } \underline{3}_\beta)$

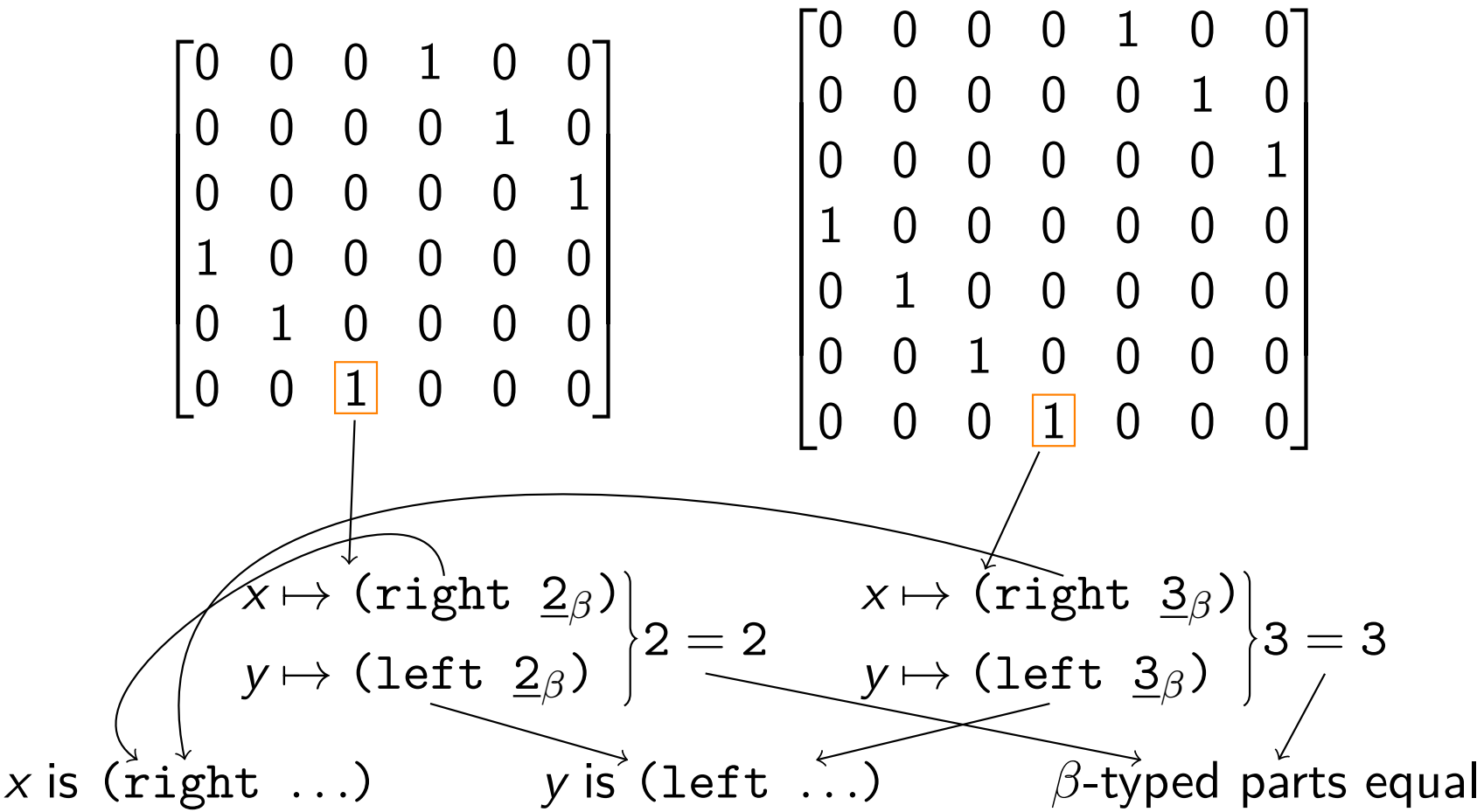
Equality patterns



Equality patterns



Equality patterns



Equality patterns

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & \boxed{0} & 1 & 0 & 0 & 0 \end{bmatrix}$$

$$x \mapsto (\texttt{right } \underline{2}_\beta)$$
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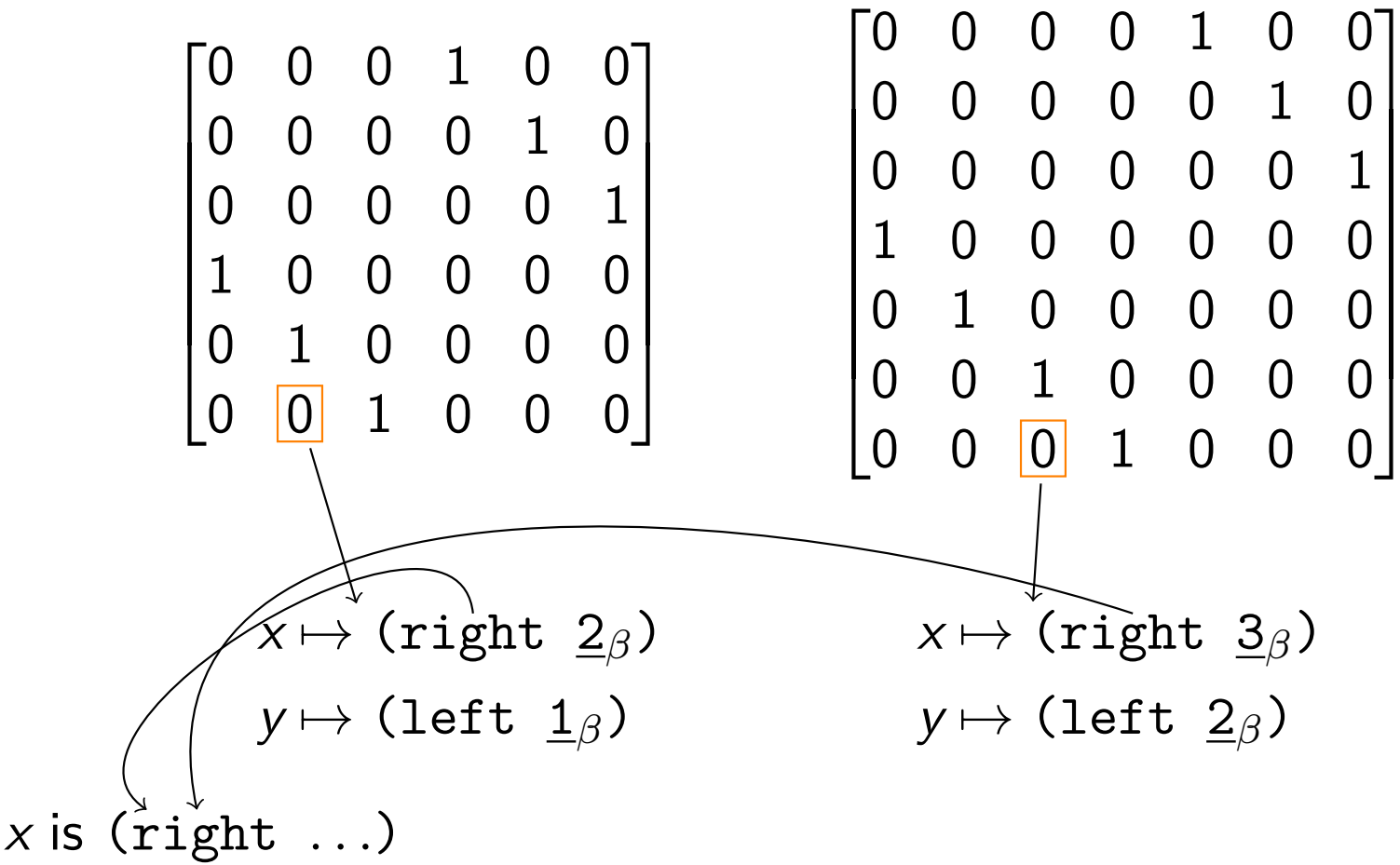
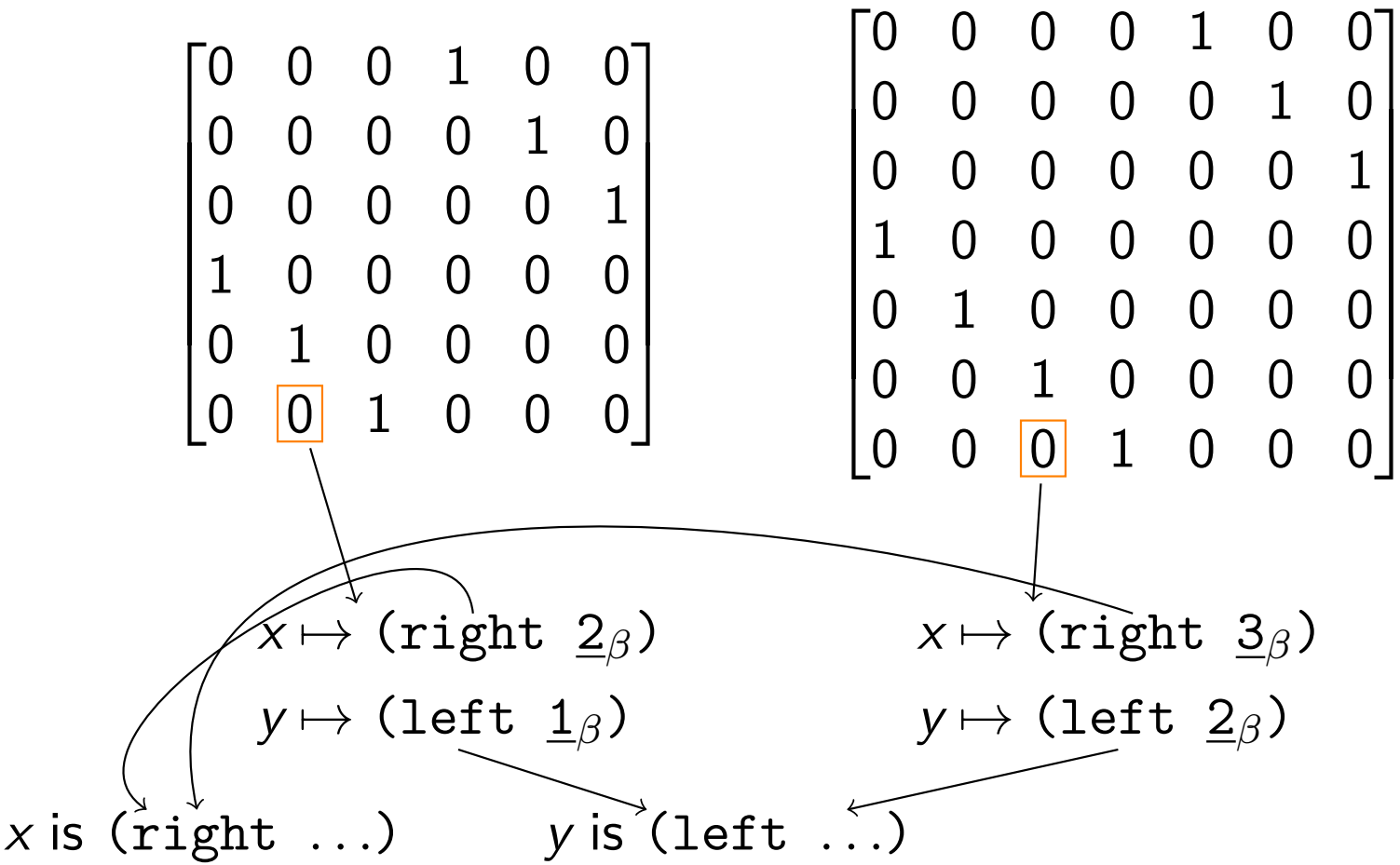
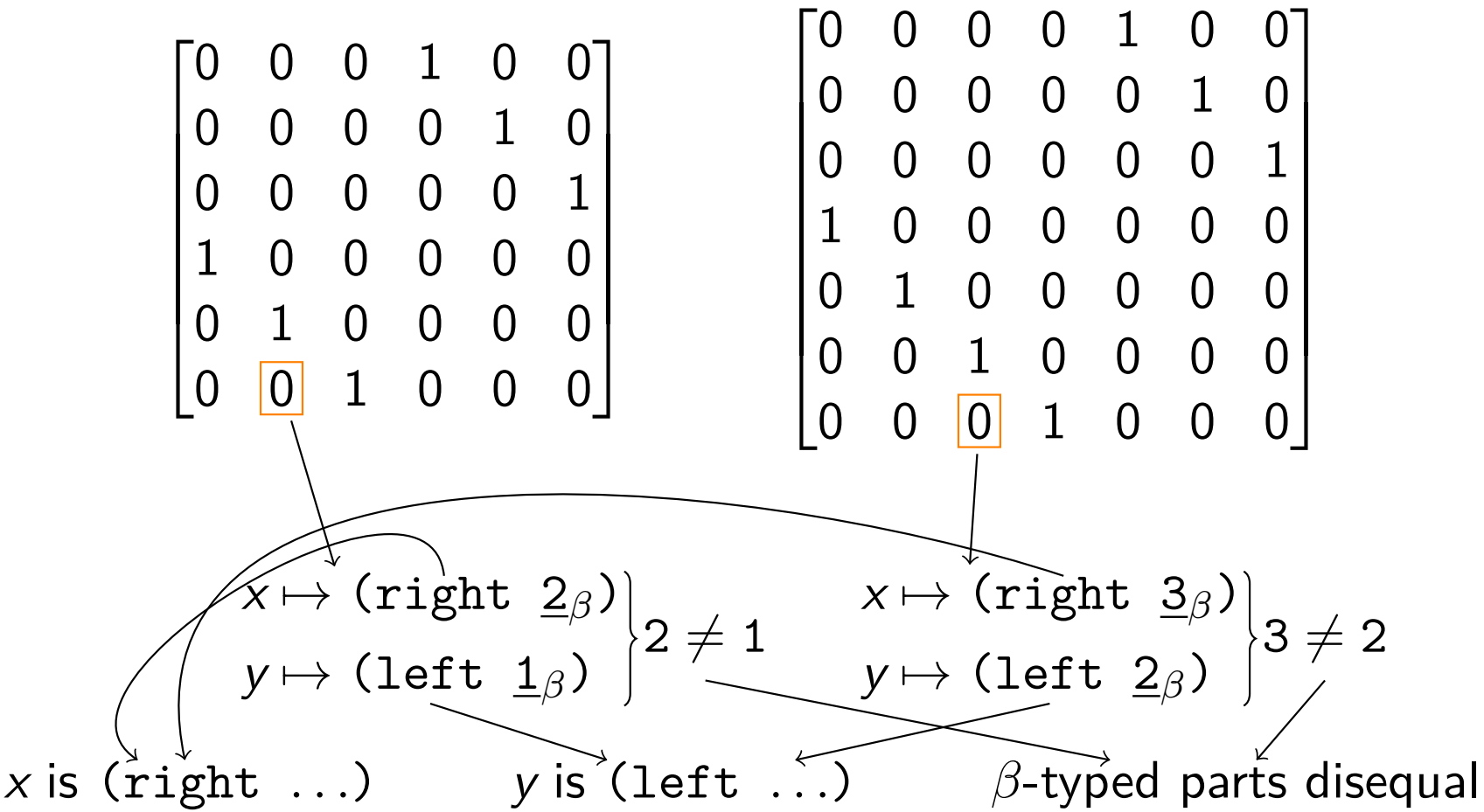


Equality patterns



Equality patterns



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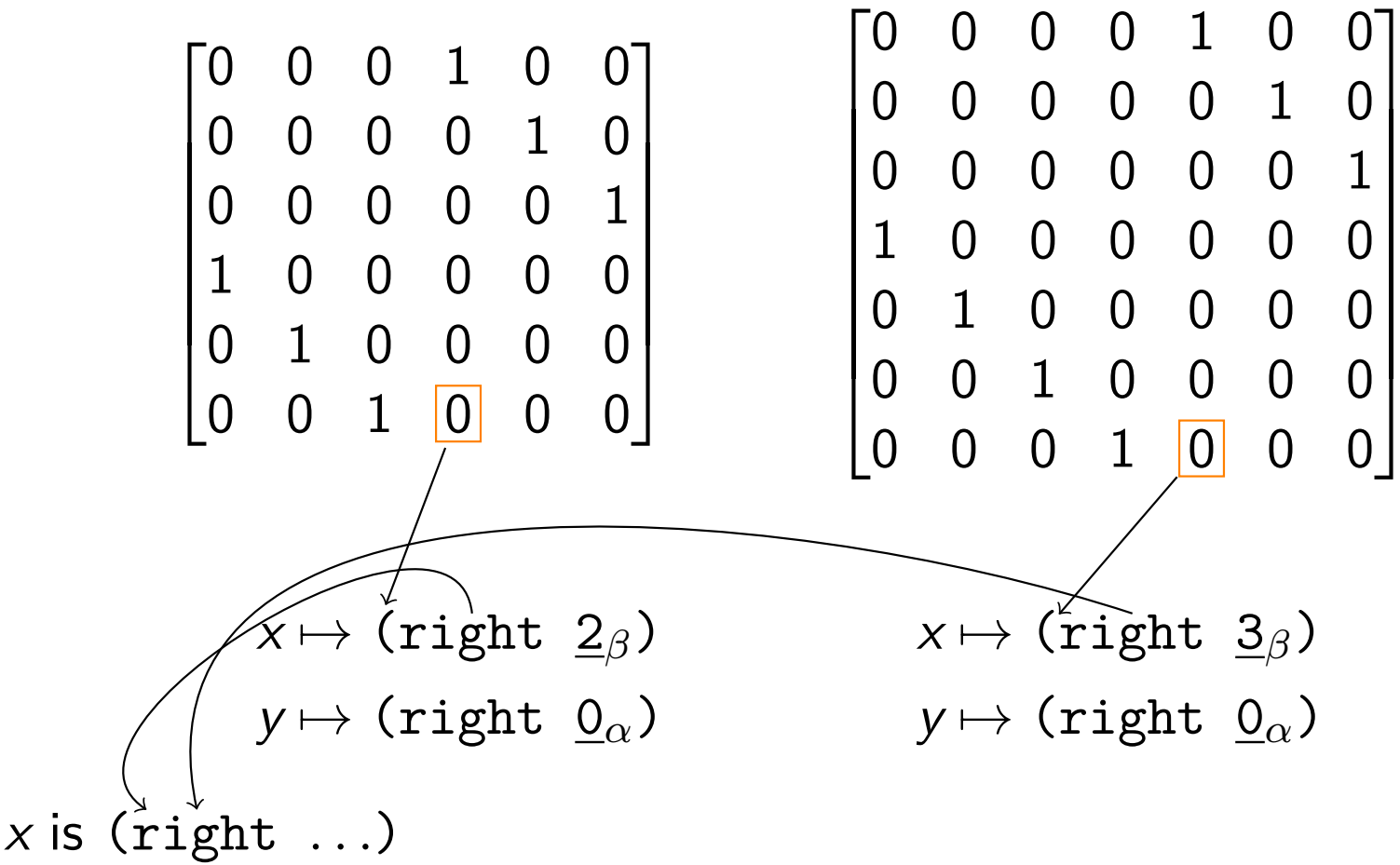

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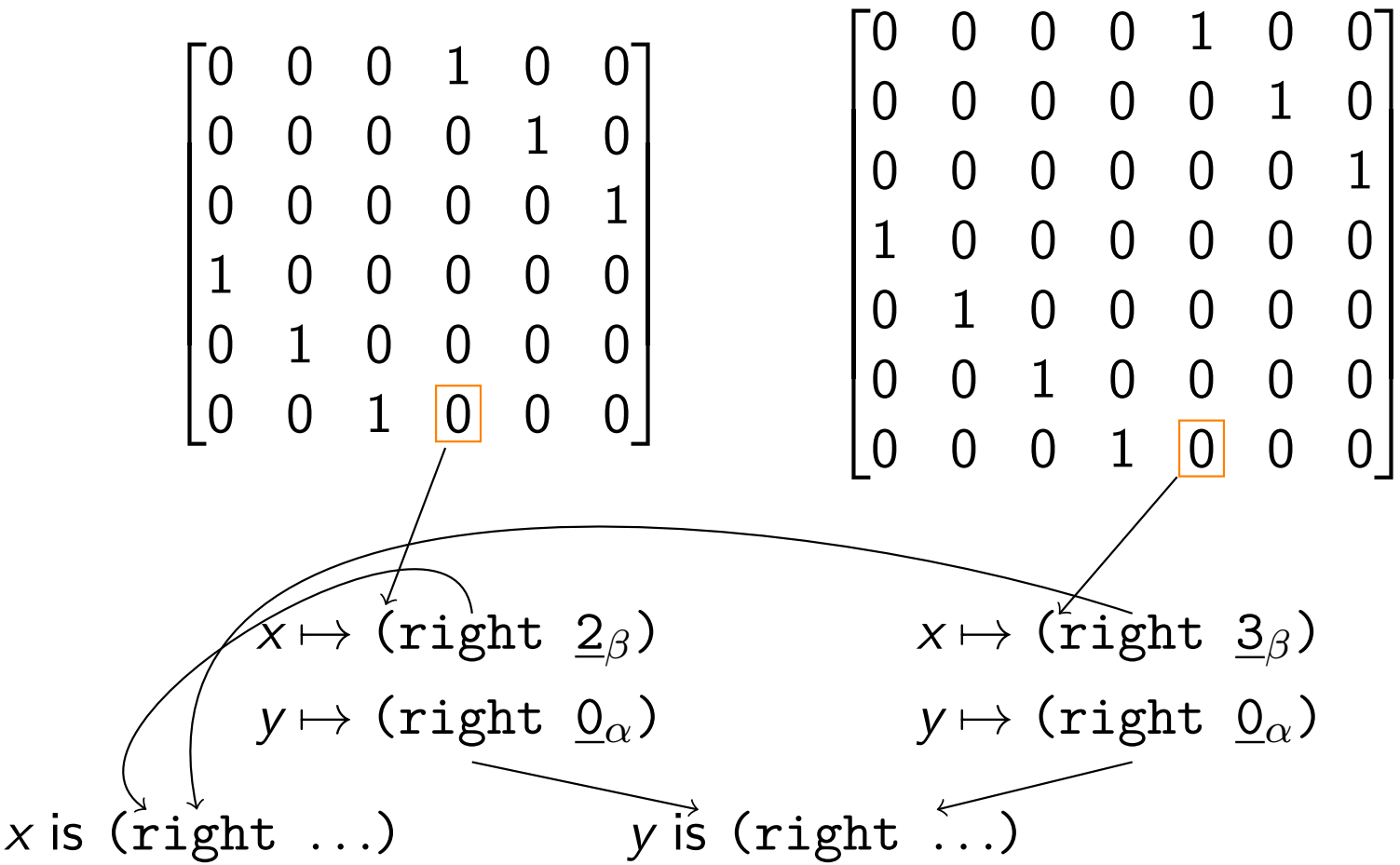

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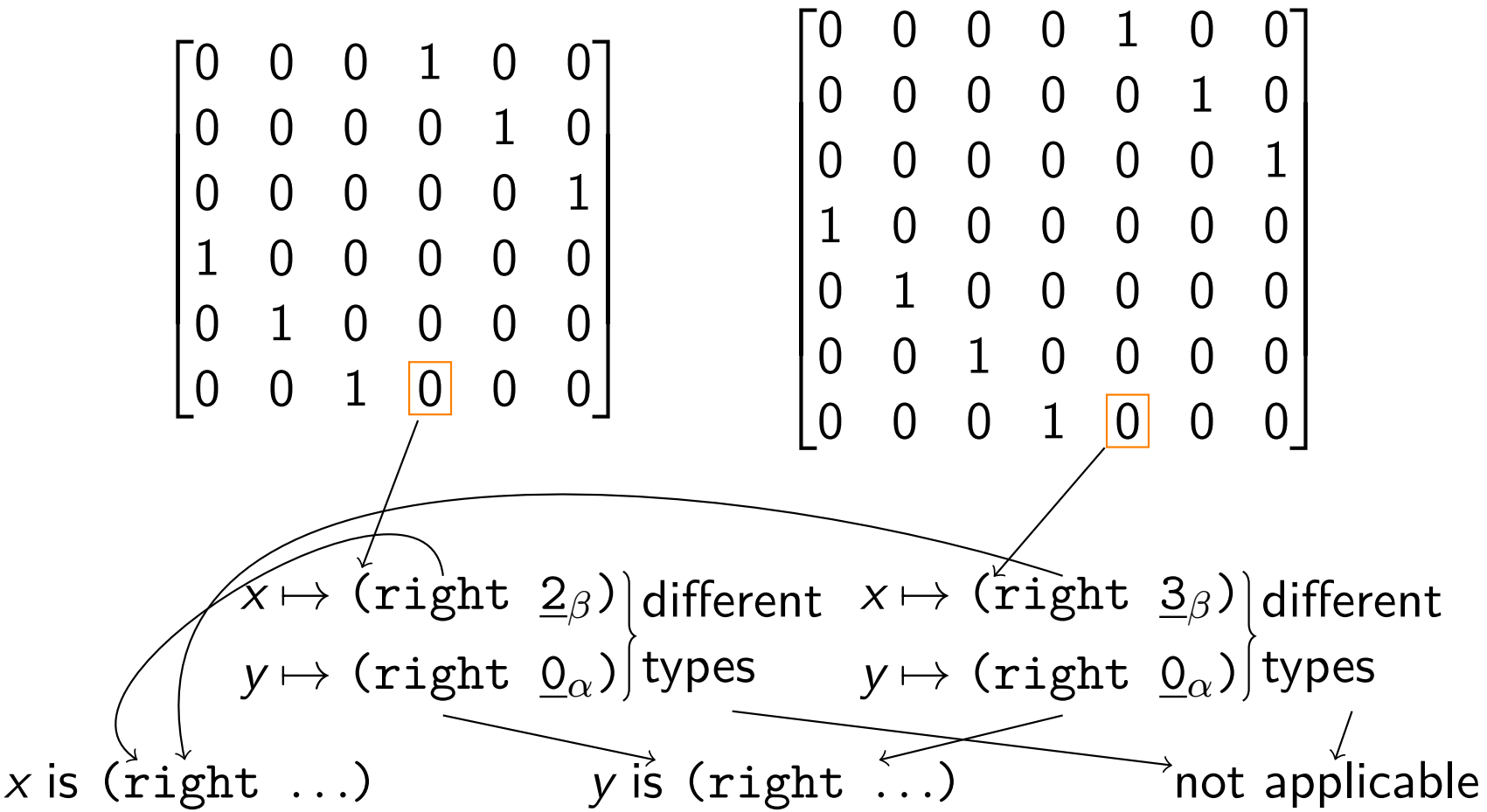

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All done?

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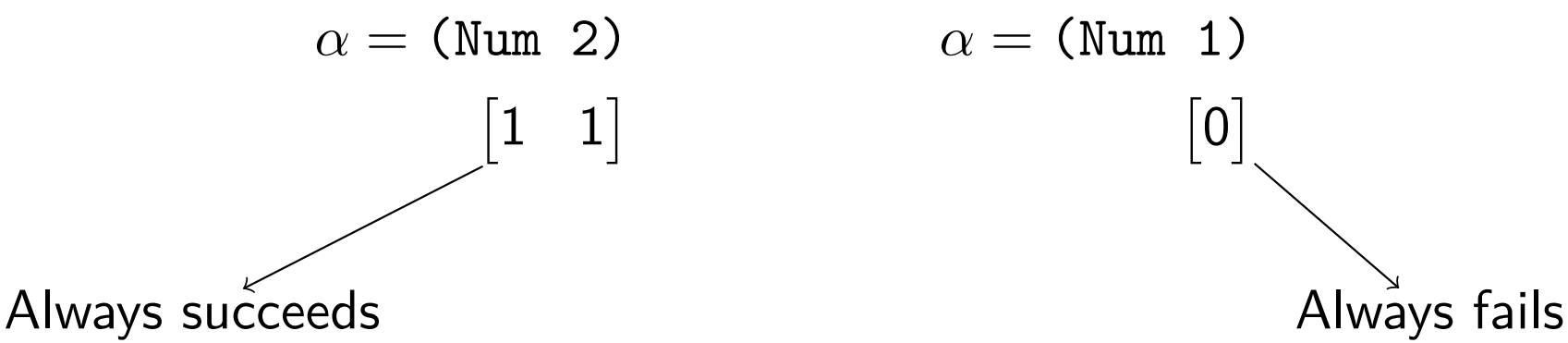

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Always fails

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Need to handle small-sized types separately (not "all for one", but almost).

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2. Replace polymorphic relation calls with concrete (monomorphic) relation calls
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More details in the paper!

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Does this generalize?

- To more than 2 arguments/array dimensions? (hint: yes)
- To recursive relations? (hint: yes)
- To more than == and =/=?
- To recursive data types?
- To non-relational languages?